

DRY-RUN GUIDE | 07_lcm.java
Source: java\2nd_belt\07_lcm.java

CORE BEHAVIOR

Computes gcd with Euclid, then evaluates $(a/\text{gcd}(a,b)) * b$. This code does not normalize signs or special-case both inputs zero.

REPRESENTATIVE INPUT / CALL

Input: '4 6'.

STEP TRACE

gcd=2; $(4/2) * 6 = 12$.

EXPECTED OUTPUT / RETURN

'12'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | Write a program to find the least common multiple (LCM) of two numbers.
004 |
005 | Input:
006 | Two integers A and B.
007 |
008 | Output:
009 | Print the LCM of A and B.
010 | */
011 |
012 | import java.util.Scanner;
013 |
014 | public class LCM {
015 |     public static int gcd(int a, int b) {
016 |         while (b != 0) {
017 |             int temp = a % b;
018 |             a = b;
019 |             b = temp;
020 |         }
021 |         return a;
022 |     }
023 |
024 |     public static void main(String[] args) {
025 |         Scanner sc = new Scanner(System.in);
026 |         int a = sc.nextInt();
027 |         int b = sc.nextInt();
028 |
029 |         int lcm = (a / gcd(a, b)) * b;
030 |         System.out.println(lcm);
031 |         sc.close();
032 |     }
033 | }
034 |
```